

AMENI KHABTHANI

AI Engineer | Voice AI Specialist

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building and deploying production-grade AI solutions for international clients, including NLP, LLM-based, and speech systems and conversational AI assistants. Covers the full lifecycle, from data preparation and model fine-tuning to API development and cloud deployment.

CORE SKILLS

AI & LLMs: Machine Learning, Deep Learning, LLMs, Generative AI, AI Agents, Prompt Engineering, Fine-Tuning (PEFT, QLoRA)

Speech & Voice AI: ASR, TTS, VAD, Speaker Diarization, OpenAI Whisper, Coqui XTTS v2, Orpheus TTS

Tools & Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, FastAPI, Flask, Supabase, Docker, Cloud services, Git

Languages: Arabic (Native), English (Professional), French (Professional)

PROFESSIONAL EXPERIENCE

Freelance AI Engineer | Remote Feb 2025 – Present

- **Custom Text-to-Speech Systems:** Fine-tuned Coqui XTTS v2 (multilingual text-to-speech) models for multiple clients across languages, accents, and voice types, including Hindi, Hinglish, Moroccan, a female Saudi voice, and children's voices; supported each project from data preparation to training and evaluation.
- **Multilingual Dubbing Automation:** Transformed a third-party dubbing workflow into a custom multilingual pipeline using Whisper Large v3 ASR, GPT-4 translation, speaker diarization, and hybrid XTTS/Orpheus TTS, deployed via Flask API on Azure App Service.
- **AI Travel Agent:** Engineered a FastAPI voice assistant for real-time hotel search and booking, integrating Whisper ASR, Groq LLMs, and TTS with intent detection, entity extraction, and conversational memory.
- **AI-Based Astrology API:** Designed and deployed an end-to-end API using LLM prompt engineering and intent classification for real-time personalized responses, with Supabase for user data.
- **Medical Translation System:** Established an English–Arabic text-to-text translation system that preserves domain-specific medical terminology with high clinical and semantic accuracy, for healthcare transcription tools.

Data Science Intern | Satoripop, Sousse, Tunisia Mar 2024 – Aug 2024

- Fine-tuned Whisper for French, English, and Tunisian dialect ASR, reducing word error rate (WER) by 45% on the target datasets.
- Fine-tuned XTTS v2 for speech synthesis, improving naturalness and intelligibility, validated through user feedback.
- Built FastAPI APIs that decreased response times by 30% and integrated with external applications.
- Developed Gradio demo interfaces to improve accessibility and simplify manual testing.

Data Science Intern | Stunas, Sousse, Tunisia Jun 2023 – Aug 2023

- Developed an OCR invoice label recognition model using DocTR for text detection and Named Entity Recognition (NER) for label extraction, reducing manual processing time by 50%.

EDUCATION

Master's Degree in Data Science, AI, and HealthTech Oct 2022 – Oct 2024
Pristini School of AI

Bachelor's Degree in Electrical Engineering (Automation and Industrial Computing) Sep 2019 – Jun 2022
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CERTIFICATIONS

- [Natural Language Processing Specialization](#) – Coursera

- [Generative AI with Large Language Models](#) – Coursera
- [Generative AI Engineering with LLMs Specialization](#) – Coursera
- [DeepLearning.AI Specialization](#) – Coursera
- [Python Data Structures](#) – Coursera
- [Data Analysis with Python](#) – Coursera
- [AWS Academy Cloud Foundations](#) – AWS Academy